ActInf Textbook Group ~ Cohort 4 ~ Meeting 4 (Chapter 1 part 2)

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all right it's our second discussion in cohort 4 of chapter one uh does anyone at the beginning want to just bring up any comment or any specific question about chapter one or otherwise we have we can look at one uh recent stream transcript okay so Terry um you asked a question about the generative model the generative process so maybe just summarize that question and then and then we'll see how it played out recently uh my question was when we were trying to create a model we create the generative model we sort of Define it in four over four sort of domains and and and then at then you define the generative process but my if if we you know think of the Markov blanket as separating B internal state from the underable external State well if we're creating a model and that model depends on us creating something that is in fact on mobile or is there not a corruption within that thought process so it creates an impression that within the context of if I take my Markov blanket is bad my skin then um I uh I cannot know what is the generative process so there's got to be some flaw in our system of modeling if it depends on creating something we can't do all right awesome and uh when you when you asked that it made me think of the recent um stream with Maxwell ramstead now operation and Ali and others and uh this highlighted a uh inconsistency in the generative model generative process uh topic Ollie do you want to summarize what you what you asked and and what they wrote or said yeah of course uh well uh we were talking about uh the I mean discrepancy between Maxwell's presentation and also the recent paper because uh in figure one in inner screen paper uh they've indicated generative model as encompassing the whole shebang I mean encompassing the internal States the external States and of course the blanket States so um nowhere generative process was visible in this diagram so we were discussing with Daniel Sanjeev and others where exactly uh General to process can be manifested uh in this diagram so I basically asked this this question where is the generative process in this diagram and uh surprisingly Maxwell uh somehow disowned the earlier literature and generative process and basically uh I mean uh he he he just I mean interpreted the generative model in a kind of new more sophisticated way that wouldn't require any separate entity as generative process because General to model basically includes the joint probability of every possible of uh every everything that markup blankets um uh I mean partitions so it it is a joint probability of internal States blanket States and the external States uh but on the other hand when we want to uh model a situation and not just I mean describe a physical phenomena or a physical system then of course we can Define our own generative model as something or let's put it this way as kind of dynamics that we expect the system or the situation the situation we have to follow those Dynamics and so for instance if we want to model a very simple linear behavior of a I don't know mechanical system then of course the Newtonian mechanics or you know the terms the lagrangian mechanics of the system can constitute the generative process of the system so uh yeah that's uh basically the uh gist of Maxwell's argument uh at least as far as I understand it quite a discussion we can kind of parse it out better but the transcripts out there like so um as with everything in the textbook it's sort of like there's a there's some internal ontology and then um it's somehow aligned with the broader active Gestalt and for that you will still it helps to understand the generating process and the particles of the agents and so um like to to the to to the earlier question how can you model the unknown because you're modeling something into existence in the act of modeling so you can have the real temperature of the room do statistics on an unobserved that maps to the observables so in practice there's no issue but then people have a lot of um degrees of interpretation with what that means and that's the whole and what that means for different situations but all that aside in terms of these two ways to talk about generative model generative process [Music] as it's being shown in the inner screen paper the generative models just the entire process giving rise to 100 of the system's level description including states that generate other observables whereas in the textbook they go to some length and distinction to highlight the generative model which has an aboutness of a generative the underlying particular partition like separating it this way is not changed so it's the same topology of the of the underlying graph that gives rise to all the you know formalisms but it's just kind of it's a truly a nomenclature question about where you say okay now these it's like what yeah it's just nomenclature question about describing this but it's one that always comes up what does anyone think about this it's kind of a funny random but cool ongoing topic uh sorry if I may add uh some additional comments to what I've just previously said I mean to my mind this recent interpretation actually uh makes more sense because uh one of the misconceptions around fep uh has always been how I mean how can we sure about how to model generative process if the causes of the observables are as the name suggests latent States or hidden States right so if we don't know uh the exact causes of those observations how is it possible to model it in a precise probabilistic probabilistic terms but in this new way of uh looking at the generative model versus generative process well we can basically I mean somehow put this issue under the Rock by by somehow reinterpreting the generative model as uh including both the external States and the internal States and only account and only following the variational density of the internal States as I mean tracing or tracking or not tracking the external States so that's basically uh I mean the whole thing the whole model and and its whole relation with the quote-unquote reality is something uh more akin to probably an instrumentalist view of reality rather than I mean the realest uh the uh perspective because uh here we just used this theoretical framework only to account for the parts of the reality or the behavior with the reality that we can it it looks like Alexa tautology but that's basically what I understood by this a new interpretation uh so that's I am probably one of the ways to evade those uh criticisms or misconceptions around the generative process thank you Ali very interesting just to kind of connect this to a previous discussion in live stream six one of the early paper-based discussions this was like described as the tale of two densities and we probably even use this meme still um but the tale of two densities was the generative direction from the temperature in the room hidden State emitting thermometer readings was the generative model a generative Direction it's referring to the same graph still and the recognition density was like from the sensor data the incoming and so that led to a lot of um well like okay the generative is action the recognition sensory and uh I think what Ali is pointing to is that if you just kind of say uh well even if it has a million nodes whether one's internal external blanket is relative to whichever one you're talking about so those are not absolute States so you might as well just call the whole thing a generative model still with certain uh components whose sparseness gives them certain properties but there's no need to necessarily distinguish between these two um but still it makes sense to think take it think about an agent-based modeling perspective and about the process that generates so so it's not that it invalidates what the textbook says just from a first thought it's it just sort of expands the scope of what people will probably mean with generative model which is just to say the whole generating architecture of the program probably also closer with generative AI so it's it's um probably a good direction to go but it will still but but this could be thought of as just like agent and Niche well can I yeah yeah this could be a problem about for me about using language right so whenever you say there is a temperature in the hidden external state when you say temperature that's a perception uh you know we can think of it as a representative of the amount of energy in that external State and we perceive it as a core layer of heat uh so so but but actually we we kind of no idea what that what that what that is we use this term temperature uh oh but we don't know what that means inter if you think of that you know like the the physics of information Pro or physics is information process and so that the temperature is information that exists in the all knowable external State we our perception of temperature is actually part of the electromagnetic spectrum you know the um infrared light is what were is what we detect as temperature and then we call we we call it a thing in the x or we may wait for we we make an assumption about what that means in the external state but really I I kind of think that we uh our our quality of temperature isn't what we mean when we measure the temperature of the external state with a thermometer or a temperature receptor in our skin you know so I again I always feel like I'm talking like I've taken too many drugs and I'm a 14 year old but you you know it it cut this this kind of it feels like there are loads of like syntaxers in the thought processes around all this so I'm I'm and um it it it confuses me Lexi yeah I was had a perspective on this question please bear with me if it sounds a little too simplistic and you know think about it this way you know I'm quoting Mark songs by the way if my body is immersed in an environment with you know 70 degrees uh Celsius temperature and I don't solve that problem I die if my body is immersed in the environment with zero degrees Celsius and I don't solve that problem I die so at some point it is not about complex philosophical things because you can go into physics originally temperature and entropy were variables designed for a system at equilibrium so we can get as semantically complex as you like but again taken back to bodies you know there is that very specific thing that if you you know are outside of the homeostatic range of temperatures and you do not solve that problem originally through reflexes through sweating or whatever and maybe through Behavior if you don't solve it you die so how is that for the you know temperature is that okay yeah but just one point on temperature I think you you um it's a great question and you you are indicating like sort of the real ontological base existence reality which of course even people have different views on that but you can model the unobserved state as like we could say this is the the um the SAT score but we're not fully observing it because we we have a partially ripped up file so the modeling of what is a hidden State versus an observable is really just a question of like what do you have data for or want it and then what do you want to include as a causal or latent factor in your model but you don't necessarily have data for as an observable but you may even choose to have hidden States for things that you could get data for so I think that's the kind of deflationary bent in this textbook and and maybe in many people's active inference Journeys when the model itself is built as they encourage people to do from the first chapter to the last then you see the specifics of how it plays out in that generative model and the contributions to the field consist of specific generative models made and general claims with the active inference ontology the thing um that it struck me is to in order to act in the world to remove myself from the temperature that's going to burn my skin I don't actually have to know anything about the generative process I can make assumptions about the generative process and then act in the world to try to alter that but my my confusion lies with how we can then within model construction dictate what the process what the generative process is and I can understand why you have to do that if you've got to create a model but it kind of just doesn't speak to the to the the ideas laid out in chapter one in this book you know that's that's another angle is uh when you approach a specific question or a line of inquiry you're not trying to make the the total um twin through a scientific model you could learn something about the knee without having the full simulation whatever that would even mean of a knee I in the specific modeling but maybe that's coming from an experimentalist's perspective so I'm not sure Alexi yeah I I had I had maybe in a reaction but I wonder if maybe we need to maybe time to maybe let this idea sink is it appropriate for me to ask another question on a slightly different topic or would that be too abruptly yeah definitely that sounds totally good if you are if it's here or just ask it so it's about chapter one and the uh I'm reacting to this text action is quintessentially goal directed and purposive okay so um I understand that you know there's a goal to stay alive no questions right but you know if we move to human subjects and you know uh uh uh please forgive me sort of Clinical Psychology Psychotherapy and there's a ton of action that humans do that at least doesn't appear as gold directed or pervasive in fact if every action humans did was goal directed and purpose if we wouldn't need Psychotherapy at all like we have I don't know maladaptive behaviors we have obsessions compulsions ticks and whatnot so uh I'm not exactly sure if we stand with that postula that every action is going to essentially goal directed and pervasive would that model not oversimplify things for human subjects that's a great question a short thought on it and then anyone can raise their hand is even an isocade can be purposeful purposive because you can explain it in terms of using this framework so it doesn't necessarily mean that in some integrated way it's beneficial for the person's life or for any given social recognition or bodily Health necessarily because you can do modeling of pathologies too but the purposiveness is just local teleology local directedness moving the elbow from one to to do a slot machine again and again is purposive you can argue whether it is with the person's long-term professional interest but the action is still proposive yeah well my my issue with the teleological explanation is that it is often uh very speculative when people for example tell me that my state of depression has a purpose to inform me that I have an unmet need which is mainstream you know models and Clinical Psychology sometimes like how do you know that depression doesn't have five different purposes and models we pretty much you know if you if you say in first speculate that that's the the purpose of depression so the teleology is is a little kind of weird for me maybe it is mainstream and active inference framework but it's a little weird yeah but maybe I'll think about it more a related question is this when we say the goal directed character of action the question is whose goal again going back to human subjects there's conscious there's unconscious processes and sometimes people do something and they're like I did this thing I rationally know I shouldn't have done it but I did it anyway which suggests that they have multiple components of their agency and not just you know a unitary John who you know decided all the goals and purposes so the question is whose goals what what exactly do we assume unitarity do you assume that the agent is a singular thing yeah great question Kate well my thoughts on this are that um I would say that in my understanding at least active inference views the Mind as a hierarchical structure but definitely hierarchical buildup or many different sub layers of the brain and ultimately the goal of therapy like in some random therapist but in some way it could be the integration of all these different sounds but those in by themselves the lower lower levels or lower parts of the hierarchical components of the hierarchy within their own domain they are purposing action the action within those domains they don't they don't solve the host necessarily as such but within their own logic internal logic even if you could put it in the math it would fit the free energy principle and the active influence so the goal there is to figure out how to bring the hierarchy together in a more purposive or healthy way that's that that's the thought that your comment triggered Alexi that's all I have to say about that I I'm very much for this thought and it's beautiful except you know bear with me maybe I'm a devil's advocate but what I see in clinical work and what is mainstream and psychodynamic Psychotherapy is mental conflicts such as I want this but I cannot have it or I I should be doing this but I want to do something else or I Want A and B and they're mutually exclusive and in that regard there is no hierarchy there is an unresolved conflict mind and a conflict is very mainstream kind of 100 year old clinical tradition and you know ultimately you don't resolve those things you know that that mind goes from one conflict to another to the third one we have multiple needs like the Famous Madonna syndrome is an example of that and how people find a compromise from that is a sort of personality so I I just do think that some of these assumptions and postulates we make early on that there's this you know either it's resolved through the hierarchy or there's a singular agent whose goals and purposes we follow are you know okay for for a bacteria but maybe not for human mind well but um card Horace Harris um his studies on psychedelics and the um his theory of how the psychedelics would kind of reset the priors reset the Precision of the priors the villains and in many cases it oftentimes helps with some of the symptoms which would argue for that view so you may not be able to do it verbally through a verbal interaction or through language but in some other you know parts of it may be open to more chemical or you know neural transmitter kind of related reset as opposed to a verbal reset but anyway just just to kind of um give it one thought and then ali um goal is not really a required term just my personal view in the active inference ontology we don't talk about kicking a ball through a goal we don't talk about a milestone based goal it's it doesn't come up when we're actually using the active infantology it's not a parameter in the model but cybernetics has long been antilios and all these debates so I think actually what's really exciting and I'll say here for now is it actually points towards exciting different ways of understanding purpose of behavior and where goals and rewards come into play by thinking about likely paths being selected rather than rewarding paths and how that generalizes across a system that stays in a wooden Groove and how that also May relate to these higher order sophisticated cognitive functions like narrative but that is not something that we're going to see in figure 4.3 that's something where once you build that Tower of Babel then you'll know but hypothetically what can you say without the generative model of what people are talking about because we're not talking about just the base four nodes modeling the whole brain like that if you were going to make a model that had multiple debating personalities then it simply would have that feature Ollie well actually I think uh the term action here is used um instead of in I mean to refer to intentional action so I I don't think it's action in its broadest sense but even if uh even if we restrict ourselves to only intentional actions there are some theories both philosophical and in cognitive science uh that tries to somehow frame this notion of action in terms of an alternative conception of causality so one notable work in that regard is the work of Alicia in her work in her book Dynamics and actions so in this book I mean the basic premise of this book is exactly uh the question what is the difference between a wink and a blink right so uh I mean in order to come in order to come up with a viable theory of action that that can include all of these uh intentional and non-intentional actions uh she proposes a theory of causation based not on uh traditional linear cause and effects but rather looking at the whole complex adaptive system as um as as possessing some constraints and looking at those constraints as uh as as somehow giving rise to causation so you know the words the what we mean by causation uh is actually undertaken by those constraints of those uh complex adaptive systems sorry so these dynamical constraints uh can actually account for both bottom-up and top down causal relations so in this sense it can include both intentional action and non-intentional action and if we take goal-directed action as uh kinds of action that tries to satisfy those dynamical constraints then this um I mean claim in the book can make sense in this complex dynamical system framework great thank you Lexi I just had a couple reactions uh Daniel to what you said and thank you for your comment about gold being necessary or not please forgive me if my metaphors are very simplistic so the car sitting in the parking lot idling burning gas is increasing entropy it is uh not moving anywhere right a car going to where I needed to go is going toward a goal and perhaps it is wasting some energy and as well so if we remove the goal out then you know how will we actually look at the free energy versus you know the energy used toward a meaningful purpose you know so I think implicitly perhaps a goal like the goal is to survive for an organism and to maintain survival evolutionarily survive until reproductive agent reproduce you know some kind of goals like that exist perhaps you know if you take all of them out then then it's all entropy right and also had a maybe reaction to Kate that you know I am very very respectful of Cartwright Harris's work and psychedelics with that I just I'm highly skeptical of the idea of reset or anything brief because you know attachment style in a human being takes from 8 to 66 to uh 18 months you know nine women pregnant for a month will not produce a baby you know personality takes time to form and all of this sort of quick you know uh resets whether it's Ayahuasca or ketamine or even electroconvulsive therapy I just don't buy that you know I think that that it's minded and conflict remains in a conflict it just how it solves the conflict is is the question it may be solving it more adaptively more you know in a more nuanced way but I don't think that anything quick like a magic wand of psychedelics will will resolve that in any way so yeah one paper by Adam saffron is the Albus model so while that relaxation of beliefs was more visible and led to a kind of um relaxation based emphasis on what psychedelics might do which may be half the picture um Adam just kind of generalized it to altering like well depending on which generative model you make or how you think about what the brain is or whatever um different um pharmaceutical or non-pharmaceutical experiences are going to not just purely relax a distribution because the distributions are part of the map so I think um like it it's not a property the physical molecule to do so except again interpreted in this more abstract causal space to your question Alexi you know you talk about a car in the parking lot it's great it's just disordering material and it's not going anywhere it's not goal oriented but its purpose could be described hyper locally as just being in one place like a rock's purpose just using again this broader concept of purpose which doesn't come up in the acting phonology we don't need the goal but the Rock's goal is just to be right there the Leaf's goal is to go Downstream so by having a deflationary optional concept of intentional Behavior including um like Carl first um wood lice example where there was insects running around on the pavement and they went faster where it was light and slower it was dark so they accumulated where it was dark so including a whole Continuum of intentionality based upon return to an attracting set is it just turns out to be enormously General now how to connect that to the the physical entropy lost by organisms open question and then how to use some other um terms from thermodynamics like what would informational work be versus informational Heat well Chris Fields is certainly approaching those types of questions but I hope that addresses a few pieces of it something doing the most likely thing is exactly the Criterion that you would use first for statistics and so in that way it's why we can use statistics so um adeptly in this modeling because statistics isn't about simply maximizing reward but like all statistics is about finding um this path of least action in an informational space Ali sorry it was my hand risen from before sorry oh no worries anyone who hasn't uh brought something up just definitely has gone all over which is totally cool but if somebody wants to go to a specific question of chapter one or a quote we could do that um I did want to chime in on the the previous topic about action is quintessentially goal directed and Professor uh so I mean you know we have some ways to get further into the textbook but I mean we're going to be going over things like policy selection kind of like I'm assuming active inference involves not just settling upon a single unchanging policy right continuing with that that's what an agent operates with rather it would be you know potential entire series of competing policies that may very well be actively updated based upon what the agent has maybe learned in the past new incoming observations so I'm just kind of reflecting on everything that's been said and um because whenever I read this I I would it feels a little bit easier for me to take uh the phrase goal directed with a bit of a grain of salt and think like oh there could be multiple potentially even conflicting policies in play that themselves are constantly potentially changing or shifting a variety of ways so just yeah it's just a kind of final comment they supposed on that topic I was thinking about yeah that's awesome it's it's so true policy is evaluated every time steps so this is exactly how we integrate action into the loop whether like ooda Loop or just behavioral uh loop so that's a great point Terry and just uh the way I interpreted that was that the goal isn't uh that the goal is fundamentally to um answer questions of the universe and uh therefore uh get evidence that then can be fed into the variational free energy model uh to either update or prioritize or act in the world to to reduce the variational free energy so the goal isn't a goal to achieve an end the goal the end is to answer the question about minimizing variational free energy and then our behavior in the world that we read as policy LED goal directed is actually more a function of that primary objective of the organism which is to minimize variational free energy great points again I'd highlight that we can Model A baseball on a parabola as doing the most likely thing we could identify a radioactive isotope that's probabilistically emitting things over a given time window as doing the most expected thing so a lot can get brought into statistics which does breed this whole fascinating area of like map territory type distinctions so suffice to say action is going to be totally integrated into this model enabling a lot of interpretations as the recent literature Witnesses Lexi thank you so much Daniel I was just yeah thinking about everyone everybody said but I think your word intentionality connected it from me and maybe the issue that I have please forgive me is lexical because so in my world you know intentionality is required the the sentence you said that Rock's goal is to stay in place is an oxymoron and and sort of psychology because you can only have a goal if you have a mind so in that regard bacteria that moves toward the like you know gradient of the food and withdraws from Pain does not have a goal and a purpose because it ain't got a mind so that that maybe is a different use of language and if they use the word gold here in a sense that the Rock's goal is to stay in place it's an attribution I would perhaps it's an unfortunate use of a term but maybe we can say It's You Know The Rock's Dynamics started to stay in place but it doesn't have any goals you know there's no mind there there's no agent there's no agent capable of choosing a versus B in Iraq yeah I I I totally agree like it is sort of a in a narrow and a broad sense sometimes there's going to be like gray area assignment just like um well this enzyme wants to bind to this one I've heard it a thousand times so it's like but it's not intended as a reflexive you know statement about the protein being a strange Loop maybe it's below some level of critical ability but we can model this all in an integrated fashion so we can still model the game theory of something that we don't even think plays it implicit representational game that's why there's this whole representationalist debate and then we can do all these kinds of modeling links really composably which is kind of the exciting piece which is also why the the kernel ultimately what we're working towards like figure 4.3 these generative models the kernels vary it's it's not applicable to any given situation except a purely purely pedagogical and it's extremely sparse on commitments because it's basically of the same type as every other statistical model in Bayesian statistics so it's not like an esoteric type of but it opens many questions that I think again we see every day which is really exciting and part of the fun and and why many people can contribute in as well but there's just so many interesting angles to it and I think also sanjeev's textbook that um is in progress with several of you um here who who may be um in that group like that there's a whole statistics degree slash undergraduate or Beyond leading up to a lot of things that really help set the stage and and his his work I I think will be like for people who want to see that level of depth and then meanwhile understanding what other um fun paths and ways are there in this whole work because certainly it's always every sentence you could be like what is happening and uh you know if this word meant that and if this person was defining it this way so certainly finding a path on through and like what's the claim when a lot of it is internally referential however that's part of its coherence and consistency and those the but sometimes the evidence for the consistency is a little bit hard to convey because these are relatively understandings hopefully that's just like a little context foreign anyone else have a thought or a question on chapter one Terry or than anyone else and it's it's relevant to this thing about that a rock has no mind but a rock guy is you know maybe over longer time scales to sustain itself as a rock it has to uh minimize its free energy in terms of its uh interactions with the external State and whenever the uh variational free energy within that system Alters it will never be oxidize or you know get broken up or if they think you know it it can't act in the world to maintain its variational free energy at a minimum but it still has to maintain that variational free energy at a minimum and when it can't it changes to something else so you know the notion that because that that that you know all objects and in the world sustain themselves and they're not in in the state that they are currently in by whether you're a tree a human a rock or a you know a Galaxy probably um uh I I what's your reaction to that because I I know that we we you know they reference like trees aren't can't do things but trees do communicate with each other you know as we and and I suppose I was obstruct by this concept of pan psychism that exists within you know the sort of the the the thinkers in in in in in in the nature of Consciousness I'm just interested yeah it's it's something that's stimulated by what Alexi said and I'm interested in what you have to say about that yeah it's very it's interesting I mean it's certainly interesting areas you brought up and the way that there could be a kind of deflationary account where just it's pan cognitivist or just pan computationalists but then people debate what's a computer and they say well it's pancognitivist but then they debate what is cognition is the baseball

Computing and it's the water Computing where it expects to be downhill so those are all the debates but most of us see a world with dense but sparse connectivities it sometimes it's like a lot of problem admiration which can be a good thing but um pointing to communication Pathways amongst living things is only surprising in light of extremely recent specific scientific observations so but I'm I'm not sure I mean that's there's so many other areas you brought up in the Consciousness and I mean these are all the huge questions Ali uh yeah again uh one of the other common misconceptions about fep is that some people think fep as providing a kind of ontological conditions for existence or at least persistence through time but actually what fep claims is that if something exists it must have properties that look as if they follow the path of least action so it's the other way around fpp doesn't provide any conditions for the Existence or persistent through time at all but it rather I mean if we take the persistence or the existence as the basic premise of our argument then we can and follow that by consider by describing the behavior of uh of the phenomena or or the system by describing uh the um I mean trajectories of their variational density uh in terms of uh least action principle so uh I think this is one of the things that people uh some sometimes criticize fep as something that I mean of course it's claimed to be theory of every thing but it's not something as some ontological or metaphysical theory for anything it just tries to describe I mean beginning from the ontological existence it tries to describe the ensuing characteristic six of the systems that persist through time and in particular highly complex high-dimensional self-organizing systems amazing points Ali thank you the condition of modeling something is to have these properties every measurement that's that's if it's static it's it's uh with its movement is within the measurement the classical limit if it's so Dynamic it's moving faster than the time scale measurement you just get basically uninformative signal so we always kind of tuned in Bucky Fuller zone of modeling Proxima modeling which we can extend with microscopes uh telescopes um that's one way to look at this that's a kind of like scientific realism approach but then for things that you can't or didn't model you have hidden States so ultimately you have it covered um but then this is really the the inversion of the um ontological theories of everything that this is how it really is this is how it is out there um this is how the metaphysics are that you know project our world a certain way that is like why ontology is the philosophy of what is however if we precondition upon existence someone will look at an and wanted to go off the trail there it's like but you have the whole evolutionary history that sets up its decision making so maybe the purposiveness of that you might have a super deflationary response um but again whatever level of sophistication the thing is from essentially inert or totally inert to repetitive motion to sophisticated motion that can be understood by an observer as a path of least action as they do statistics so in that sense it licenses little more metaphysics than a linear regression as an analytic method then I think again there's so many interesting angles why it has been um discussed in all of these more philosophical contexts because reductionism isn't the whole picture so the theory in itself or any given theory in itself might not take on all these accounts and there may just be tangled webs of perspectives and and difference languaging and uh different applications around some of these topics but it is fun and fast when it's small and the few people's perspectives are essentially knowable and there are um always new information coming in literally through these funny modern channels and hearing what Maxwell and Carl at all think Andrew yeah yeah I just uh just my two cents there is that um you know I find the pan cognitive stuff um rather interesting than that said I personally came to active inference because I'm interested in its application to uh living systems and especially human beings and so the brain uh from a neuroscience perspective but also how it might be applied and working with psychopathologies treatments and so on and so um I think so far I'm just viewing it in more pragmatic sense of like how can we use this in the same way that I also work as a data scientist and so I don't view everything as a gradient boosting machine but I will attempt to apply a model to a situation to see if I can get anything useful out of it right so that's that's kind of how I've taken this so far I mean I've read some really fascinating papers you know one by a psychiatrist who applied active inference to uh post-traumatic stress disorder described um hyper precise priors in the upper cortical regions is like you know someone who experiences reliving the um excuse me that can be triggered by a certain phenomena that kind of spark or activate those those previous priors that gets superimposed on the incoming sensory observations even though it's inaccurate um so that's I mean I'm personally a little less concerned with the Precision of ontological claims and so on but um but but kind of how this can help us to reinterpret different kinds of phenomena and from there maybe use it like for good for some kind of useful purpose and sounds really it's really fascinating you know so it's awesome thank you for sharing that it is very interesting anyone else want to just give a comment in the last minutes or otherwise we will be going into chapter two chapter two is gonna be uh another two weeks we can look at the figure and look at the high road and low road again just kind of look ahead low road and the high road chapter 2 and chapter three that will be the coming month also some people have asked about the math learning group I I made a little comment thread if you just comment on this thread then I think people who want to do a certain kind of work can maybe find each other this way and it can be in the Discord or a gypsy link or however people prefer yeah continue to to check out and comment or or improve if you want any of the AI math things as we head into chapter two explore which of these equations if you can evaluate which one of these AI explanations or make comments on them some are useful some are not so you know but I hope there's a lot of fun things we can do in the coming weeks chapter 2 is definitely going to get into it because it just starts the real non-introduction part of the book but yeah does anyone have any other thoughts or questions or anyone who's read chapter two before Wanna Give a teaser or or Ali go for it well yeah actually chapter two is where the real fun begins I mean after the introductory chapter and now we're uh into the real set of things by considering the two of the most Central equations of active inference namely equations 2.5 and 2.6 one for variational free energy the other four expected free energy which will come to time and time again throughout the whole book so it's probably one of the most fundamentally important chapters great comments anyone else want to add anything okay all right thank you all see you around bye